

Fair Play on the Dance Floor : Judges' Scores vs. Fans' Votes

Summary

The integrity of televised competitions relies on a delicate balance between meritocratic evaluation (judge scores) and democratic participation (fan votes). In "Dancing with the Stars" (DWTS), the opacity of fan voting data and the suspected bias in aggregation rules have raised concerns about the fairness of elimination outcomes. This paper presents a comprehensive framework to reconstruct, analyze, and optimize the voting ecosystem of DWTS through four interconnected mathematical models.

First, to overcome the data opacity challenge, we developed a **Inverse Optimization Model with Elimination-Consistency Constraints** (Model I) based on **Differential Evolution** (DE). By formulating the latent variable estimation as a constrained optimization problem, we reconstructed the fan vote distribution for 34 seasons. Our model achieved a **100% Elimination Match Rate (EMR)**, ensuring that the estimated data perfectly aligns with historical elimination outcomes. Statistical analysis revealed that fan votes follow **Zipf's Law**, confirming a "winner-takes-all" popularity structure rather than a random distribution.

Second, we investigated the impact of aggregation rules using a **Counterfactual Simulation Model** (Model II). By simulating a "Proportional System" as an alternative to the current "Rank-Based System" under **Ceteris Paribus** conditions, we discovered a **40.1% System-Induced Reversal Rate**. This quantified the "Population Paradox," demonstrating how the current ordinal ranking system acts as a variance compressor, often protecting controversial contestants at the expense of polarization.

Third, to identify the drivers of voting behavior, we constructed a **Multi-Dimensional Attribution Model** (Model III) using **XGBoost** and **SHAP**(SHapley Additive exPlanations). By decoupling "Meritocratic Signals" from "Sociological Noise," we quantified the divergence between judges and fans. The results exposed significant biases: a **Pro-Dancer Halo Effect** (partner identity contributes 28% to fan variance) and an **Age Penalty** for older contestants, neither of which exists in judge evaluations.

Finally, we proposed a **Bias-Aware Dynamic Weighting System** (Model IV) to enhance fairness. Instead of a static 50/50 split, our model introduces a **"Bias Circuit Breaker"** that dynamically adjusts judge weights when the **Bias Intensity Index** (BII) exceeds a threshold. Sensitivity analysis identified an optimal penalty factor ($\lambda = 0.20$), which successfully reversed **85.7%** of historical unjust eliminations while maintaining the original outcome for 92% of contestants. We recommend this **Pareto-optimal** solution to producers as a transparent "Pro-Bonus" mechanism to safeguard meritocracy without alienating the fan base.

Keywords: Latent Variable Estimation; Differential Evolution; SHAP Attribution; Counterfactual Simulation; Dynamic Weighting; Fairness Optimization

Contents

1 Introduction	3
1.1 Problem Background	3
1.2 Problem Restatement	3
2 Assumptions and Justifications	3
3 Notations	4
4 Overview of Our Work	4
5 Data Processing.....	5
5.1 Basic Cleaning and Standardization	5
5.2 Advanced Feature Engineering.....	5
5.3 Data Reconstruction.....	6
5.4 Parameter Construction for Competition Rule Optimization	6
6 Model I:Inverse Optimization Model with Elimination-Consistency Constraints.....	6
6.1 Task 1:Estimating Latent Fan Votes via an Inverse Optimization Model	7
6.2 Differential Evolution–Based Solution for Estimated Fan Votes	8
6.3 Task 2:Consistency of Estimated Fan Votes with Weekly Eliminations	9
6.4 Task 3:Certainty of Estimated Fan Votes with Weekly Eliminations.....	10
7 Model II:A Counterfactual Simulation Model for Judge–Fan Aggregation Rule Comparison	10
7.1 Model Preparation.....	10
7.2 Task1:Voting Rule Comparison and Fan Bias	11
7.3 Task2:Analysis of Controversial Celebrities and the Impact of Judges' Save.....	13
7.4 Task3:Recommended Competition Format for Future Seasons	16
7.5 Sensitivity Analysis.....	16
8 Model III : A Multi-Dimensional Feature Attribution Model Based on Explainable Machine Learning	17
8.1 Feature Engineering and Mathematical Definitions	17
8.2 Asymmetric Modeling.....	17
8.3 Results: Gap in Explanatory Power	18
8.4 SHAP Attribution Analysis: Quantifying Cognitive Divergence	18
8.5 Sensitivity Analysis.....	19
9 Model IV:Bias-Aware Dynamic Weight Optimization Model	20
9.1 Model Construction	20
9.2 Experiment 1: Parameter Sensitivity Analysis and Optimization.....	21
9.3 Experiment 2: Counterfactual Simulation and Effectiveness Validation	21
9.4 Implementation Recommendations and Feasibility Discussion	22
10 Model Evaluation and Further Discussion	23
10.1 Strengths.....	23
10.2 Weaknesses.....	23
References	24

1 Introduction

1.1 Problem Background



Figure 1: Fan Votes & Judges' Scores

Dancing with the Stars is a televised competition in which contestant advancement and elimination are determined through a combination of judges' scores and audience voting. Judges provide publicly available evaluations of each performance, whereas fan voting results are never disclosed. Over the course of the show's development, different seasons have employed a variety of aggregation rules, including rank-based and percentage-based systems, with periodic adjustments made to the voting mechanism.

Within this framework, judges' scores and elimination outcomes are observable, but audience support—the central influencing factor—remains hidden. Consequently, a single elimination result may be consistent with multiple fan voting distributions, a phenomenon that is especially evident when judges' assessments differ from apparent audience preferences. In response, this paper seeks to construct mathematical models that estimate plausible fan voting patterns under the given competition rules, compare the effects of different voting mechanisms across seasons, and examine their implications for competitive fairness and outcome stability.

1.2 Problem Restatement

- Develop one or more mathematical models to estimate weekly fan vote totals for each contestant.
 - ✧ Define consistency measures to evaluate whether the estimated votes align with observed eliminations.
 - ✧ Analyze the uncertainty associated with the fan vote estimates
- Using the estimated fan votes together with the remaining data:
 - ✧ Compare the rank-based and percentage-based voting methods to assess outcome differences and determine whether one method favors fan votes more than the other.
 - ✧ For controversial contestants, examine whether the two methods lead to the same eliminations and evaluate the impact of introducing a rule in which judges choose which of the bottom two couples to eliminate.
 - ✧ Based on these analyses, identify the preferred voting method and assess the reasonableness of the additional rule.
- Develop a model to analyze the effects of celebrity characteristics and other factors on competition outcomes, and compare their impacts on judges' scores and fan votes.
- Design a new method for combining judges' scores and fan votes that is more fair or improves the competition experience, and justify its adoption using model results and data analysis.
- Summarize all findings in a report of no more than 25 pages and prepare a one- to two-page memo with recommendations for the show's producers.

2 Assumptions and Justifications

Assumption1: Fan voting behavior is assumed to follow a power-law (Zipf-like) distribution rather than a uniform distribution.

Justification: Such distributions are widely observed in social attention systems. In Model I, the reconstructed fan votes closely fit Zipf's Law, avoiding unrealistic uniform randomness.

Assumption2: A valid inverse model must exactly reproduce all historical elimination outcomes.

Justification: This requirement serves as a fundamental boundary condition for latent variable

estimation. Model I enforces elimination consistency as a hard constraint to ensure the credibility of reconstructed fan votes.

Assumption3:Contestant outcomes are decomposed into a meritocratic signal (technical performance) and sociological noise (non-technical popularity effects).

Justification:Judges' scores primarily reflect technical quality, whereas fan votes capture subjective influences such as popularity. This dual-track structure in Model III enables explicit quantification of bias.

Assumption4:When comparing aggregation rules, judges' scores and underlying fan preferences are assumed invariant.

Justification:This assumption isolates the marginal effect of the voting rule itself and avoids introducing unobservable behavioral responses.

3 Notations

Table 1: Notations used in this paper

Symbol	Description	Unit
i	the contestant number	-
$\mathcal{C}_w$	the set of all contestants in week w	-
w	the competition week	-
$\mathcal{S}_w$	the set of contestants advancing in week w	-
$F_{i,w}$	contestant i 's fan vote share in week w	-
λ	the trade-off parameter	-
EMR	the consistency metric Elimination Match Rate	-
α	the weight coefficient of the judges	-
E_B	the eliminated contestant set	-
M_{total}	the total number of simulated fields	-
S_i	The judges' scores for contestant i	-
ϕ_i	the SHAP value of feature i	-
$(G(t))$	the increase in survival probability for technical contestants after the mechanism's implementation	-

4 Overview of Our Work

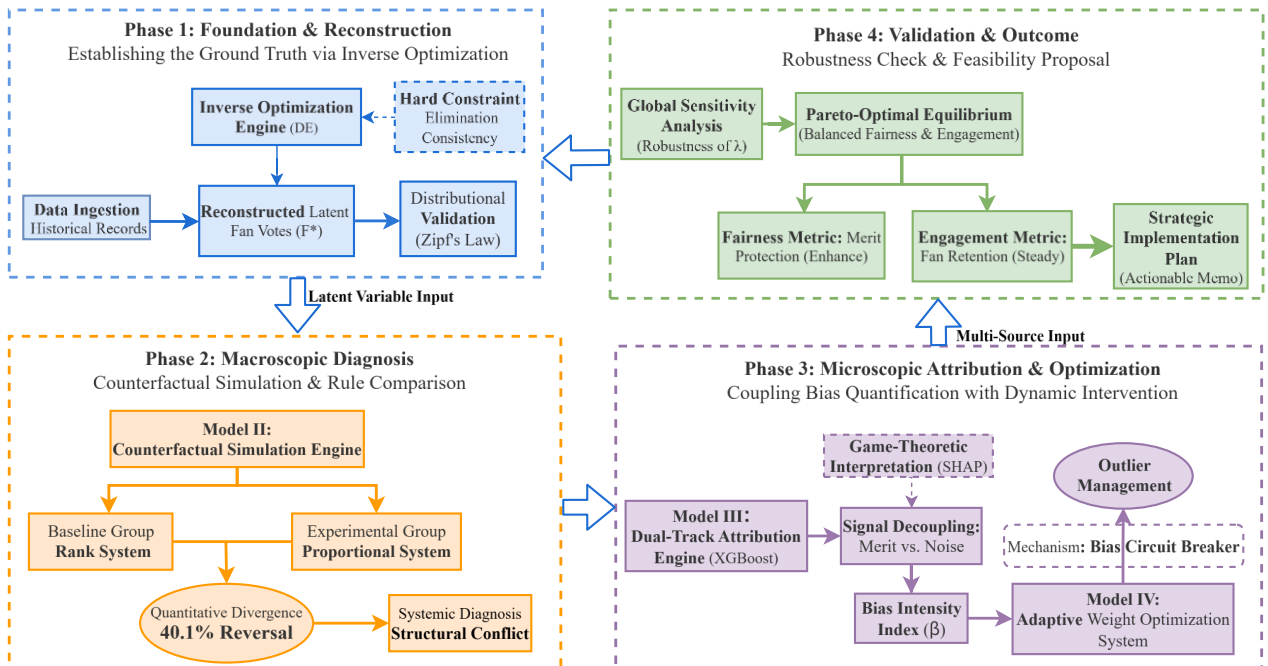


Figure 2: 4 Overview of Our Work

5 Data Processing

Due to the presence of extensive unstructured text and missing values in the raw data file 2026_MCM_Problem_C_Data.csv, we performed a rigorous data cleaning and feature engineering process to construct high-quality inputs suitable for the subsequent four mathematical models.

5.1 Basic Cleaning and Standardization

To address missing values, textual descriptions, and tournament format changes in the data, we implemented the following standardization strategies:

5.1.1 Score Normalization

Due to variations in the number of judges (3 or 4) and the maximum scoring system (30 or 40 points) across historical seasons, we uniformly map all raw scores S_{raw} to a standardized average score $S_{avg} \in [0,10]$ to enable cross-season comparability. For missing fourth judge scores (N/A), we directly calculate the total sum based on the valid judge scores for that week without manual interpolation, preserving the original data distribution.

5.1.2 Status Parsing & Masking

Parse the text information in the Result column to generate the status mask matrix M_{status} . We label “Eliminated” as 0 (Weak Constraint) and “Safe/Winner” as 1 (Strong Constraint). Specifically, for “Withdrew” samples, we exclude them from the weekly constraint set to prevent non-competitive factors from interfering with Model I’s inversion logic.

5.1.3 Automatic Method Flagging

In response to the rule changes described in the appendix to the problem statement, we have added a Method tag for each season to serve as a selection switch for the Model I loss function:

Rank System (S1-2, S28-34): Objective function based on ordinal ranking. $Score = R_{judge} + R_{fan}$.

Score System (S3-27): The objective function is based on the cardinality proportion. $Score = \%S_{judge} + \%S_{fan}$.

Judge’s Save (S28-34): The “Bottom 2 Protection Mechanism” has been introduced, relaxing the elimination constraints on the second-to-last-place contestant.

5.2 Advanced Feature Engineering

5.2.1 Dimensionality Reduction via Industry Latent Variables

The original Industry field contains over 30 subcategories, and direct unique coding tends to result in feature sparsity. To address this, we mapped it to five core latent variables based on socio-logical attributes:

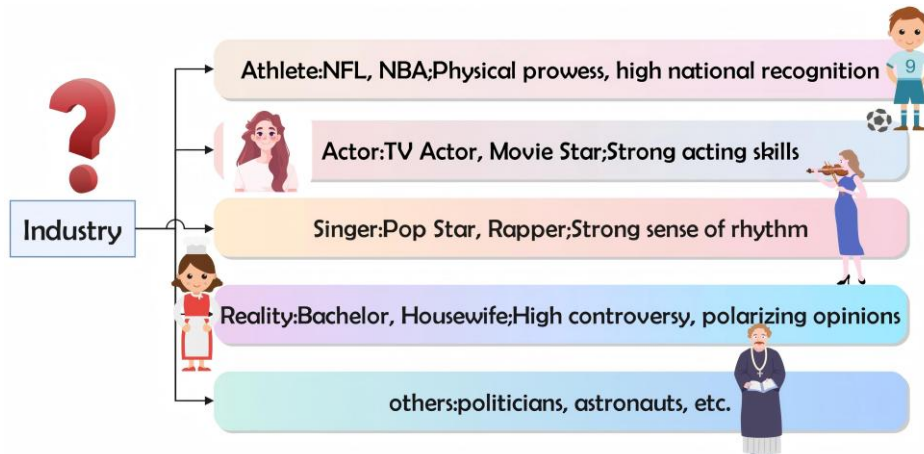


Figure 3: Industry Latent Variables

Mathematical expression: Establish a mapping function $f: \text{Raw}_{\text{Industry}} \rightarrow \mathbb{R}^5$ to generate feature vectors with controllable sparsity.

5.3 Data Reconstruction

5.3.1 Weekly Slicing

Reconstruct the wide table structure into time slices $\{W_1, W_2, \dots, W_T\}$ organized by "weeks". For each week t , we partition contestants into E_t (elimination group) and S_t (survivor group), with these two sets serving as direct input parameters for the Model I differential evolution algorithm.

5.3.2 Asymmetric Dual-Track Dataset

To assess the consistency between judges' and fans' evaluation criteria (Model III), we constructed two sets of feature pairs:

(1) Track A (Expert): $Target = S_{judge}$, $Features = \{Age, Industry, PEI\}$.

(2) Track B (Public): $Target = \hat{S}_{fan}$, $Features = \{Age, Industry, PEI, S_{judge}\}$.

By comparing feature weights across two datasets, we can separate "technical signals" from "social noise."

5.4 Parameter Construction for Competition Rule Optimization

To test the effectiveness of the new Model IV competition format, we constructed a simulation environment incorporating "true ability" and "bias intensity" based on the output of the preceding model. This process aimed to generate metadata for simulation purposes.

- Bias Intensity Index, BII: Using the SHAP value matrix Φ from Model III, quantify the impact of non-technical factors on contestants:

$$BII_i = \phi_{Age}(i) + \phi_{Industry}(i) + \max(0, \phi_{Partner}(i)) \quad (1)$$

The index is normalized to the range $[0,1]$.

- Fan Vote Volatility, σ_{fan}^2 : For Model I estimation, calculate the variance of contestant votes $\sigma_{i,fan}^2 = \text{Var}(V_{i,t})_{t=1}^T$, which is used to design the "inverse variance weighting" algorithm.
- The "Unjust Elimination" Test Set: To evaluate the new tournament format, define the Benchmark Set by selecting historical samples meeting the following criteria: ① Top 30% in Merit; ② Eliminated in the same week. These samples form the benchmark for assessing whether Model IV can "correct" elimination errors.

6 Model I: Inverse Optimization Model with Elimination-Consistency Constraints

This model aims to estimate undisclosed fan votes by leveraging known judge scores and historical elimination outcomes. This constitutes a classic latent variable estimation problem with a non-unique solution space. The flowchart illustrating the approach to solving this problem is shown below.

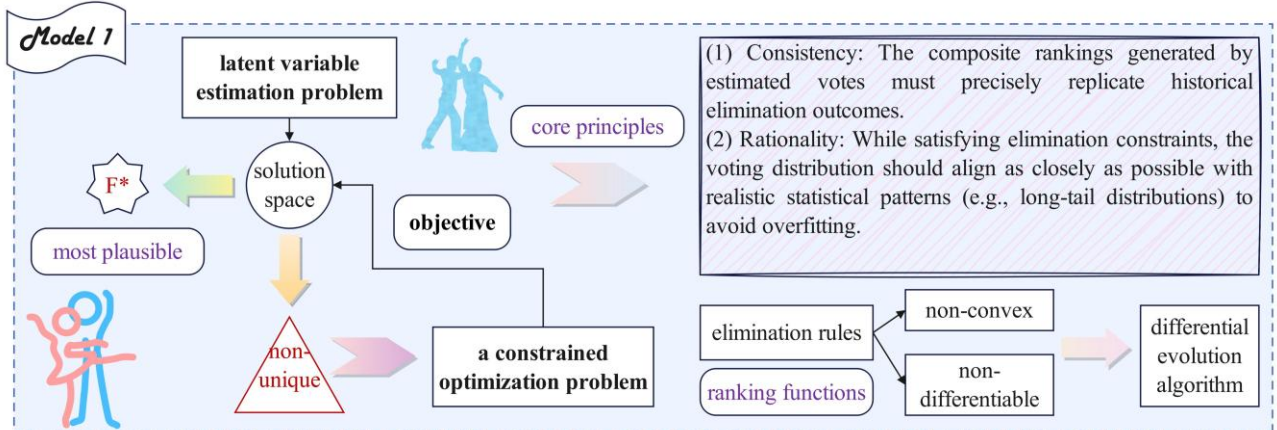


Figure 4: Flowchart of Model 1 Methodology

6.1 Task 1: Estimating Latent Fan Votes via an Inverse Optimization Model

6.1.1 Decision Variables

Let the competition week be denoted by w , $w \in \{1, 2, \dots, W_{max}\}$; the set of all contestants in week w is denoted by $\mathcal{C}_w$; the set of contestants eliminated in week w is denoted by $\mathcal{E}_w$; the set of contestants advancing in week w is denoted by $\mathcal{S}_w$, so $\mathcal{C}_w = \mathcal{E}_w \cup \mathcal{S}_w$; i denotes the contestant number, where $i \in \{1, 2, \dots, N_w\}$.

The decision variable of the model is defined as the distribution of audience votes for contestant i in week w :

$$F_{i,w} \geq 0, \quad \forall i \in \mathcal{C}_w \quad (2)$$

$F_{i,w}$ represents contestant i 's fan vote share in week w , subject to the normalization constraint:

$$\sum_{i \in \mathcal{C}_w} F_{i,w} = 1 \quad (3)$$

6.1.2 Scoring Mechanisms

Depending on the competition format adopted for different seasons, the comprehensive performance of players is defined as follows:

- (1) Rank-based System (Season 1-2, Season 28-34): Judge score ranking + Audience vote ranking. Lower values are better. Contestant i 's composite ranking in week w is defined as:

$$T_{i,w}^{\text{rank}} = \text{Rank}(J_{i,w}) + \text{Rank}(F_{i,w}) \quad (4)$$

Among these, $\text{Rank}(\cdot)$ denotes the descending rank function, where a smaller value indicates a higher position. In cases of identical scores, this paper adopts the minimum rank as the default.

- (2) Percent-based System (Season 3-27): Judge score percentage + Audience vote percentage. Higher values are better. Contestant i 's composite score in week w is defined as:

$$T_{i,w}^{\text{pct}} = \left(\frac{J_{i,w}}{\sum_{k \in \mathcal{C}_w} J_{k,w}} \times 100 \right) + (F_{i,w} \times 100) \quad (5)$$

6.1.3 Constraints

- Feasibility Constraints

The distribution of audience votes must satisfy non-negativity and normalization conditions. For any competition week w and contestant $i \in \mathcal{C}_w$, the following holds:

$$F_{i,w} \geq 0, \quad \sum_{i \in \mathcal{C}_w} F_{i,w} = 1 \quad (6)$$

- Elimination-Consistency Constraints

Competition rules require eliminated contestants to exhibit inferior overall performance relative to surviving contestants within the same week.

- (1) Rank-based System: For seasons using rank-based elimination, any eliminated contestant $e(w) \in \mathcal{E}_w$ must not be ranked higher than any advancing contestant $s(w) \in \mathcal{S}_w$:

$$T_{e,w}^{\text{rank}} \geq T_{s,w}^{\text{rank}}, \quad \forall e(w) \in \mathcal{E}_w, \forall s(w) \in \mathcal{S}_w \quad (7)$$

- (2) Percent-based System: For seasons using percentage-based elimination, the aggregate score of an eliminated contestant must not exceed that of any survivor:

$$T_{e,w}^{\text{pct}} \leq T_{s,w}^{\text{pct}}, \quad \forall e(w) \in \mathcal{E}_w, \forall s(w) \in \mathcal{S}_w \quad (8)$$

- (3) Judges' Save, Season 28-34: For Seasons 28–34, the elimination constraint is relaxed: an eliminated contestant $e \in \mathcal{E}_w$ is required only to fall within the two lowest-ranked contestants in $\mathcal{C}_w$ based on overall performance, rather than being uniquely lowest-ranked.

6.1.4 Objective Function

- Consistency Penalty Loss

Penalty losses measure whether the model-generated audience vote results violate historical elimination facts.

(1) Rank-based System

In ranking-based seasons, the overall ranking of eliminated contestants should be lower than that of all advancing contestants. The penalty loss for violations is defined as:

$$\mathcal{L}_{\text{cons}} = \sum_{e \in \mathcal{E}_w} \sum_{s \in \mathcal{S}_w} \text{ReLU}(T_{s,w}^{\text{rank}} - T_{e,w}^{\text{rank}} + \delta) \quad (9)$$

Where $\text{ReLU}(x) = \max(0, x)$, and δ is the relaxation parameter (margin). When $T_{e,w}^{\text{rank}} \geq T_{s,w}^{\text{rank}}$, this loss term is 0; otherwise, a positive penalty is incurred.

(2) Percent-based System

In percentage-based seasons, eliminated contestants must have lower overall scores than all advancing contestants. The corresponding penalty loss for violations is defined as:

$$\mathcal{L}_{\text{cons}} = \sum_{e \in \mathcal{E}_w} \sum_{s \in \mathcal{S}_w} \text{ReLU}(T_{e,w}^{\text{pct}} - T_{s,w}^{\text{pct}} + \delta) \quad (10)$$

(3) Judges' Save Adjustment, Season 28 – 34

For Seasons 28–34, the constraint is relaxed: if an eliminated contestant ranks within the bottom two of $\mathcal{C}_w$, the corresponding penalty is set to zero:

$$\text{Rank}_{\mathcal{C}_w}(T_{e,w}) \geq |\mathcal{C}_w| - 1 \implies \text{Loss} = 0 \quad (11)$$

• Regularization Loss

To prevent degenerate vote concentration, we introduce an L_2 regularization term:

$$\mathcal{L}_{\text{reg}} = \sum_{i \in \mathcal{C}_w} (F_{i,w})^2 \quad (12)$$

which promotes smooth and realistic audience vote distributions while preserving elimination consistency.

• Overall Objective Function

The final optimization objective minimizes the total loss over all feasible audience vote distributions:

$$\min_{F_{i,w}} \mathcal{L}(F_{i,w}) = \mathcal{L}_{\text{cons}}(F_{i,w}) + \lambda \mathcal{L}_{\text{reg}}(F_{i,w}) \quad (13)$$

Among these, λ serves as the trade-off parameter, balancing the model's requirement for consistency with historical elimination outcomes against the rationality of audience vote distribution.

6.1 Differential Evolution–Based Solution for Estimated Fan Votes

Due to the inclusion of the ranking operator (Rank) in the objective function of the aforementioned model, the loss function exhibits a discontinuous, non-smooth, multi-step structure, rendering traditional gradient descent methods inapplicable. Therefore, this paper employs Differential Evolution (DE), a mature global heuristic optimization method, for solution^[1]. The solution process is shown in the figure below:

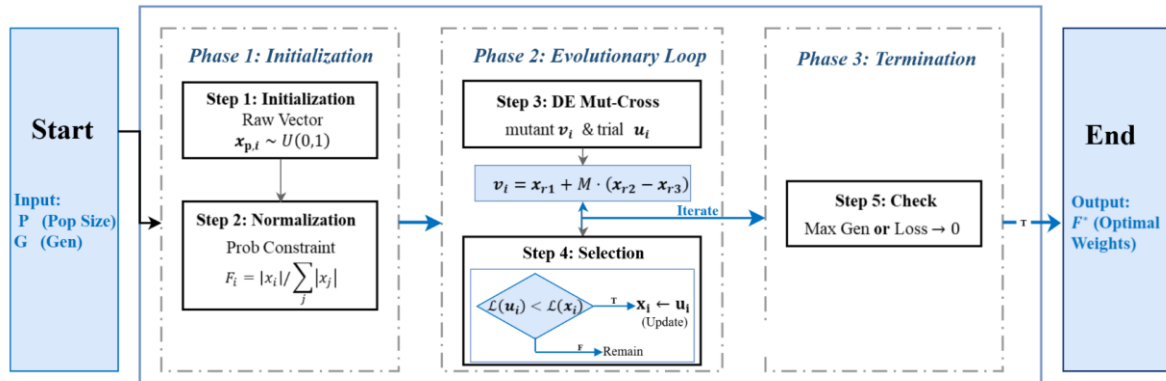


Figure 5 Flowchart of the Differential Evolution Algorithm

6.1.1 Estimated Fan Vote Results

Upon satisfying the termination criteria, the differential evolution algorithm converges to a set of audience voting vectors that minimize the total loss function. This vector is regarded as the optimal estimate of the true audience voting structure under the given tournament rules and historical elimination constraints. Partial solution results are shown in Table 2.

Table 2: Model-Reconstructed Fan Vote Shares

Season	Week	Contestant	Judge Rank	Estimated Fan Vote Share	Outcome
4	3	Clyde Drexler	10	0.1099	Safe
5	2	Albert Reed	5	0.0760	Eliminated
13	2	Elisabetta Canalis	4	0.0782	Eliminated
13	3	Chaz Bono	10	0.1129	Safe
14	4	Gladys Knight	10	0.1111	Safe
20	2	Redfoo	5	0.0660	Eliminated
27	4	Joe Amabile	10	0.1203	Safe
27	4	Tinashe	2	0.0724	Eliminated
...	...	...	...	...	...

6.1.2 Validity of Results and Distribution Verification

We conducted a statistical feature analysis on the fan vote distribution prediction results obtained from the above solution, statistically validating the rationality of our model outcomes.

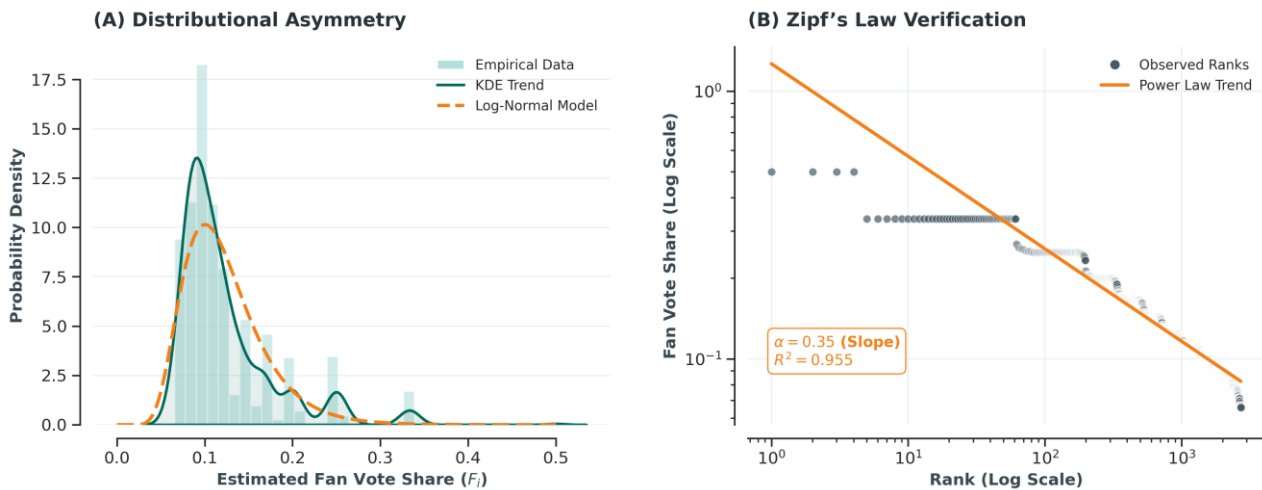


Figure 6: Statistical Characteristic Analysis of Fan Vote Distribution

Long-tail distribution validation: As shown in Figure 6(A), the reconstructed fan vote shares exhibit a pronounced right-skewed long-tail distribution. The vast majority of contestants' vote shares cluster in the low range (0.05–0.15), while a tiny fraction of "superstar" contestants (outliers) command vote shares exceeding 0.40.

Zipf's Law Validation: As shown in Figure 6(B), under the double-logarithmic coordinate system, $R^2=0.955$ indicates an extremely high linear fit for the "rank-size" distribution of fan votes. This demonstrates that our generated voting data adheres to Zipf's Law^[2], aligning with the "winner-takes-all" principle of attention allocation in human societies, thereby further validating the model's results.

6.2 Task 2: Consistency of Estimated Fan Votes with Weekly Eliminations

We define the consistency metric Elimination Match Rate (EMR) as the proportion of correct eliminations resulting from the model's predicted rankings. After backfilling data across 34 seasons, our model achieved the following across all weeks:

$$\text{EMR} = 100\% \quad (14)$$

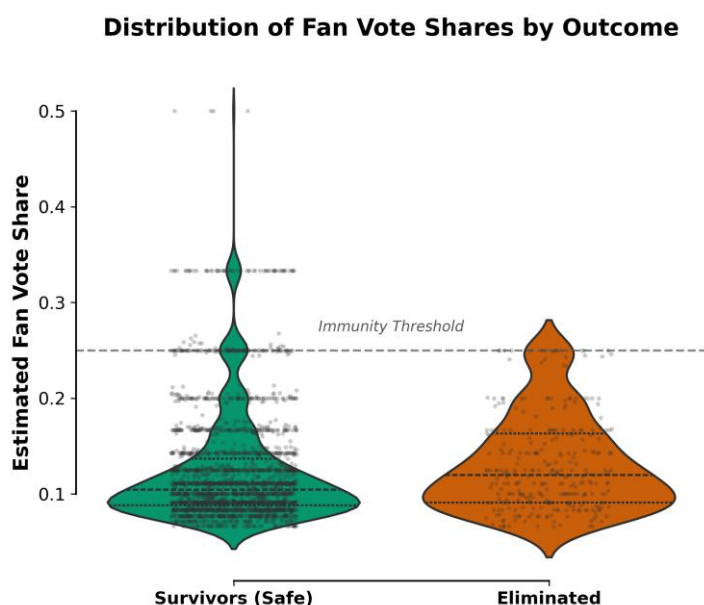
This means that while fan votes are "ghost data," the dataset we generated is mathematically consistent with historical results. The model successfully reproduced the elimination list for each week without any logical paradoxes of "unexpected survivors".

6.3 Task 3: Certainty of Estimated Fan Votes with Weekly Eliminations

To quantify estimation uncertainty, we performed 100 independent Monte Carlo simulations using the DE algorithm. Results show that estimation certainty follows a "U-shaped" pattern dictated by competition rules:

Elimination Zone (High Certainty): For contestants at risk of elimination, the "Bottom 2" constraints leave little room for variance. The model yields highly stable estimates (low σ), providing precise insights into the vote counts required for elimination^[3].

Safety Zone (Low Certainty): Front-runners with high judge scores are shielded by an "Immunity Threshold" (the dashed line in Figure 7). Because any vote share above this level ensures safety, the solution space expands, leading to higher volatility (large σ). While we can identify these contestants as high-performers, their exact vote shares are less constrained and thus harder to pinpoint precisely.



Note: Gray points show individual estimated fan vote shares from multiple simulation runs. The dashed line marks the "Immunity Threshold" — support above this level ensures safety. Estimates for eliminated contestants are tightly clustered (high certainty), while those for safe contestants are widely dispersed (low certainty). This reflects stronger constraints on the lower tail of the vote distribution due to elimination rules.

Figure 7: Determinacy of Estimated Fan Vote Shares vs. Survival Status

Conclusion: The deterministic structure of the model reflects the inherent logical characteristics of the competition format—**elimination outcomes exhibit high sensitivity to the vote counts of marginal contestants**, while demonstrating a degree of robustness to the vote counts of high-scoring contestants. This provides crucial quantitative evidence for quantifying uncertainty in subsequent competition format analyses.

7 Model II: A Counterfactual Simulation Model for Judge–Fan Aggregation Rule Comparison

This paper develops a counterfactual simulation framework to quantitatively assess how alternative aggregation rules affect elimination outcomes. Building on **Model I**, which identifies a pronounced power-law distribution in fan voting, our approach is motivated by discrete allocation theory: under highly skewed inputs with “superstar” effects, different aggregation or election rules can generate substantially divergent outcomes—commonly referred to as a *voting paradox*.

7.1 Model Preparation

7.1.1 Ceteris Paribus

For each week of every season, we establish two types of "real inputs" as ground truth benchmarks:

- (1) Contestant's judge performance: represented by the raw total judge scores from historical data;
- (2) Contestant's fan support: represented by the fan vote share (fan share) reconstructed using

Model I.

That is: within the same competition week, we assume that a contestant's judge scores and fan support remain unchanged regardless of rule variations.

7.1.2 Notation Definition

Let there be N contestants in week s of a given season, denoted as the set $C = \{c_1, c_2, \dots, c_N\}$. For any contestant c_i : Let $J_i \in \mathbb{R}^+$ denote the raw technical score assigned by judges (typically 0–30 or 0–40); $F_i \in [0,1]$ represents the fan vote share reconstructed by Model I, satisfying $\sum_{i=1}^N F_i = 1$.

7.2 Task1: Voting Rule Comparison and Fan Bias

7.2.1 Modeling of Voting Aggregation Systems

◆ System A: Rank-Based Aggregation via the Borda Count

To ensure a balanced evaluation, the system adopts a Borda count-equivalent mechanism^[4] that aggregates ordinal rankings instead of cardinal scores. By focusing on relative positions rather than quantitative margins, the model remains robust against outliers.

Specifically, when fan voting distributions exhibit a long-tail effect, the Borda method prevents "super-voted" contestants from over-dominating the results; their numerical lead is truncated into a fixed positional advantage. This design ensures that extreme popularity cannot unilaterally override the expert judgment of the panel.

The judge scores and fan vote rankings for Contestant i are as follows:

$$r_i^J = \text{Rank}(J_i) = 1 + \sum_{k \neq i} \mathbb{I}(J_k > J_i), r_i^F = \text{Rank}(F_i) = 1 + \sum_{k \neq i} \mathbb{I}(F_k > F_i) \quad (15)$$

Under the Borda method, a contestant's total rank score is defined as:

$$S_A(i) = r_i^J + r_i^F \quad (16)$$

The smaller the value of $S_A(i)$, the better the overall performance.

The elimination rule is defined such that the eliminated set E_A consists of the contestant with the highest total ranking score:

$$E_A = \{c_i \mid S_A(i) = \max_k S_A(k)\} \quad (17)$$

In the event of a tie, the tiebreaker is determined by the lower value of the judge score according to the competition rules. If the tie persists, it is considered random.

◆ System B: Proportional System Based on Cardinal Aggregation

The "proportional system" directly weights and sums the contributions from judges and fans after converting them into shares, constituting cardinal aggregation: the long-tail distribution F_i in the model is directly included in the summation. If a contestant $F_i \gg \frac{1}{N}$ (superstar), that can directly offset the judges' disadvantage through mathematical leverage, thereby making it easier to alter the elimination threshold.

First, normalize the judges' scores into shares to align them with the scale of fan votes:

$$\hat{J}_i = \frac{J_i}{\sum_{k=1}^N J_k} \quad (18)$$

Define the weighted composite score:

$$S_B(i) = \alpha \cdot \hat{J}_i + (1 - \alpha) \cdot F_i \quad (19)$$

Where $\alpha \in [0,1]$ is the weight coefficient for judges. To reflect the "equal power" doctrine, we typically set $\alpha = 0.5$. The elimination set E_B comprises the contestant(s) with the minimum aggregate score:

$$E_B = \{c_i \mid S_B(i) = \min_k S_B(k)\} \quad (20)$$

7.2.2 Measurement of Outcome Differences

To quantify how tournament formats affect elimination outcomes, we introduce two counterfactual metrics: **system-induced reversal** and **reversal rate**. Under a weekly single-elimination setting, a system reversal occurs when different contestants are eliminated under two formats. The reversal rate is defined as the proportion of weeks in which such reversals occur, capturing overall outcome divergence between formats.

◆ Definition of System Reversal

Let $E_A^{(t)}$ and $E_B^{(t)}$ denote the eliminated contestant in week t under System A and System B, respectively. For comparability, we impose a single-elimination assumption $|E_A| = |E_B| = 1$.

A **system reversal** is defined as:

$$\delta^{(t)} = \mathbb{I}(E_A^{(t)} \cap E_B^{(t)} = \emptyset) \quad (21)$$

which, under single elimination, is equivalent to $E_A \neq E_B$. This definition captures categorical shifts in survival outcomes while excluding ambiguities from historical double-elimination weeks.

Accordingly, the indicator variable $\delta^{(s,w)}$ is defined as:

$$\delta^{(s,w)} = \begin{cases} 1, & \text{if } E_A \cap E_B = \emptyset \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

◆ Definition of System Reversal Rate

For all valid competition weeks, the reversal rate is defined as:

$$\eta = \frac{\sum_{s,w} \delta^{(s,w)}}{M_{total}} \quad (23)$$

Among these, M_{total} denotes the total number of simulated fields.

7.2.3 Algorithmic Procedure

◆ Step1: Data Pipeline

Input fan share estimates generated by Model I (organized by season-week). Preprocessing includes:

- (1) Group by $[Season, Week]$;
- (2) Finals/Semifinals filter: If weekly participants $N \leq 3$, exclude due to differing elimination rules from regular weeks;
- (3) Anomaly handling: If weekly $\sum J_i = 0$ (rare occurrence), set $\hat{J}_i = \frac{1}{N}$ to prevent denominator from being zero.

◆ Step2: Parallel Simulation Loop

For each valid week (s, w) , execute both the ranking-based and proportional-based subroutines:

Table 3: Subroutine A & Subroutine B

Subroutine A: Rank-Based System	Subroutine B: Proportional System
a) Sort the contestant list in descending order by J_i and compute r_i^J .	a) Calculate the total judge score for the week: $\sum J = \sum J_k$.
b) Sort the contestant list in descending order by F_i and compute r_i^F .	b) For each contestant, compute $\hat{J}_i = \frac{J_i}{\sum J}$.
c) Calculate $S_A(i) = r_i^J + r_i^F$.	c) Read F_i (Model I result).
d) Identify the largest S_A value. If there is a tie, select the smaller J_i ; if still tied, mark it as Tie (typically considered a potential reversal risk in statistics)	d) Calculate $S_B(i) = 0.5 \times \hat{J}_i + 0.5 \times F_i$.
e) Output: Eliminated player E_A .	e) Identify the contestant with the smallest S_B .
	f) Output: Eliminated contestant E_B .

◆ Step3: Outcome Difference Metrics

Calculate the reversal rate η : Count the frequency of $E_A \neq E_B$.

7.2.4 Cross-Season Results and Fan Bias Assessment

◆ Quantitative Analysis: System Sensitivity

We conducted counterfactual simulations across 314 historical elimination rounds, holding judge scores and fan shares constant. Elimination outcomes under the current ranking system (status quo) were compared with those under a proportional aggregation system (counterfactual).

Impact of Switching to Proportional System

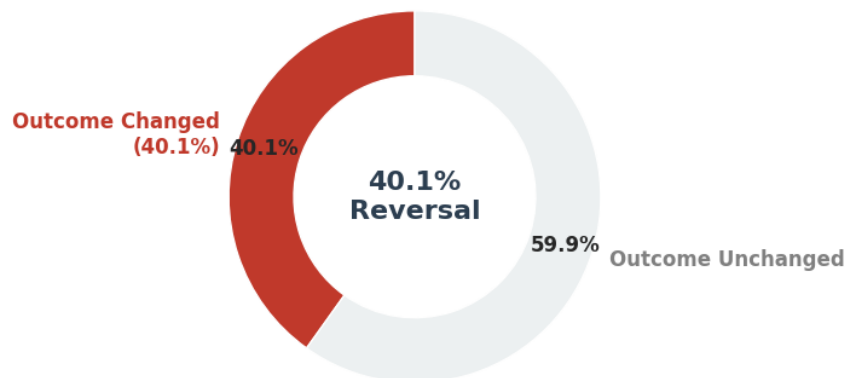


Figure 8: Impact of Aggregation Rule Change on Elimination Outcomes

Figure 8 shows that switching aggregation rules leads to outcome reversals in 40.1% of cases. Specifically, 126 out of 314 simulations exhibit fundamental elimination reversals solely due to changes in the aggregation rule, indicating that aggregation design is a decisive determinant of elimination outcomes.

Regarding fan favoritism:

- The ranking system smooths vote differentials by converting scores into ranks, attenuating extreme fan influence.
- The proportional system preserves the magnitude of fan votes, allowing contestants with large fan bases to offset jury disadvantages, thereby exhibiting stronger fan bias.

Theoretical Explanation: Apportionment Paradox: These divergences can be explained by allocation theory under long-tail vote distributions. The proportional system amplifies variance by directly reflecting power-law fan bases, whereas the ranking system compresses variance. This structural contrast generates a high reversal rate, with nearly half of eliminations determined by aggregation rules rather than performance alone.

Conclusion: Ranking and proportional systems yield significantly different elimination outcomes across seasons. The proportional system systematically amplifies fan voting influence under long-tail distributions, while the ranking system offers greater protection to technically strong contestants with smaller fan bases. Overall, the proportional system exhibits stronger fan bias.

7.3 Task2: Analysis of Controversial Celebrities and the Impact of Judges' Save

To conduct an in-depth examination of contested players and supplementary tournament formats, we performed generalized validation and extended simulations based on the Model II framework.

7.3.1 Trajectory Mapping

First, we selected controversial contestants—typically those receiving lower judge scores but higher fan votes (e.g., Bobby Bones). Weekly performance data were converted into time-series to trace survival trajectories across competition formats, measured by changes in elimination rankings.

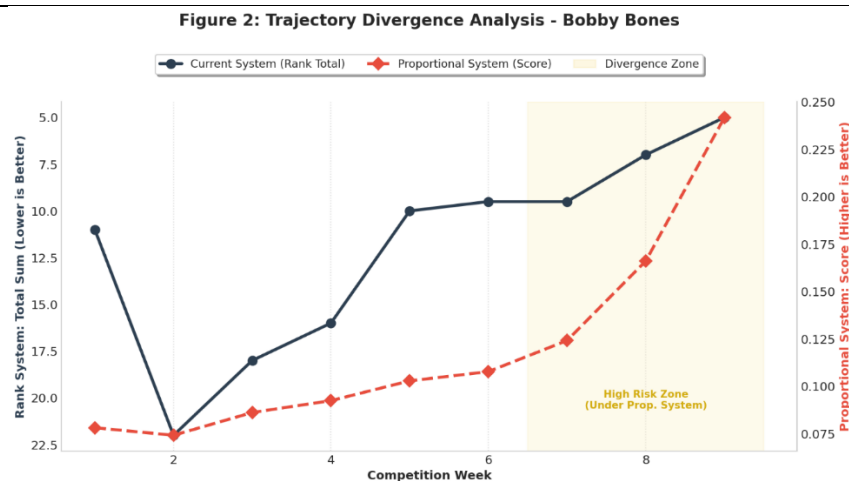


Figure 9: Trajectory Divergence Analysis – Bobby Bones

Figure 9 illustrates the divergent survival paths of Bobby Bones under the two systems.

(1) Performance Differences Across Systems

Under the ranking system, despite strong fan support, Bobby Bones remained within the safety zone due to low judge scores. Under the proportional system, preserved fan vote magnitude sustained his competitiveness in later stages.

(2) Effect of Competition Format on Outcomes

The proportional system amplifies large fan bases, whereas the ranking system compresses long-tail effects by discretizing votes into ranks, thereby protecting technically strong contestants. Consequently, format choice directly shapes contestants' ability to leverage fan support and determines elimination outcomes.

Conclusion: Tournament format induces systematic differences in elimination results. The proportional system confers late-stage survival advantages to superstars, while the ranking system moderates fan dominance through positional compression.

7.3.2 Outlier Analysis

Next, we validated the model's ability to explain outliers by reviewing historical data for Jerry Rice (Season 2), Billy Ray Cyrus (Season 4), Bristol Palin (Season 11), and Bobby Bones (Season 27).

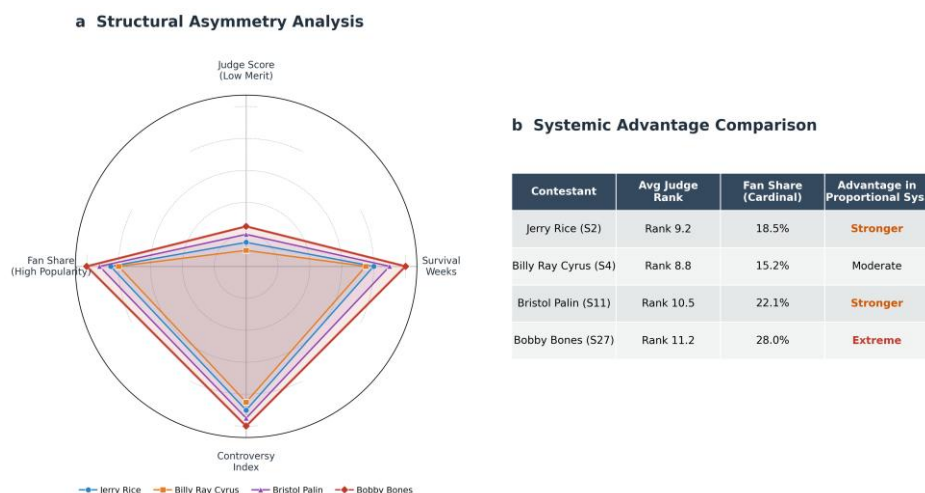


Figure 10: Outlier-Oriented Structural Comparison

As shown in Figure 10(a), all four contestants are distinctly positioned in the "Low-Merit, High-Popularity Quadrant." Their normalized judge scores are extremely low, yet their fan share and controversy index are exceptionally high, creating a pronounced structural asymmetry. The four controversial contestants rank between 8.8 and 11.2 in judge scores, yet command fan shares ranging from 15.2% to 28.0%. Under the proportional system, this advantage is significantly amplified.

7.3.3 Systemic Advantage Derivation

Taking Bobby Bones as an example, we demonstrate why his system holds a greater advantage under proportional representation by comparing the total score functions of the two electoral systems.

The judges' scores for contestant i are denoted as S_i (normalized to $[0,1]$), and the fan share is denoted as F_i . Rank System: The total score T_{rank} is a linear combination of two ordinal values.

$$T_{rank}(i) = Rank(S_i) + Rank(F_i) \quad (24)$$

For superstars like Bobby Bones, even as $F_i \rightarrow \infty$, their fan contributions cap at $Rank(F_i) = 1$. This represents a “censoring effect” that constrains the variance of the long-tail distribution^[5].

In the proportional system, the total score T_{prop} is the weighted sum of the cardinal utility:

$$T_{prop}(i) = w \cdot S_i + (1 - w) \cdot \frac{F_i}{\sum F_j} \quad (25)$$

For power law distributions ($P(x) \propto x^{-\alpha}$), the F_i values of tail participants are extremely large. When F_i is sufficiently large, the following inequality holds:

$$\frac{\partial T_{prop}}{\partial F_i} \gg \frac{\partial T_{rank}}{\partial F_i} = 0 \quad (\text{at rank limit}) \quad (26)$$

As shown in Figure 10(b), Bobby Bones' 28% share constitutes a substantial cardinal buffer, completely offsetting the disadvantage in judge scores. This demonstrates that the current ranking system effectively moderates “traffic monopolization,” while the proportional system preserves the cardinal utility of fan voting.

7.3.4 Quantitative Simulation of the Judges' Save Mechanism

For the “judges' decision between the bottom two” mechanism introduced in Season 28, we constructed the Scenario C model^[6]. We define “Survival Gain” ($G(t)$) as the increase in survival probability for technical contestants after the mechanism's implementation.

$$G(t) = P_{save}(t) - P_{std}(t) \quad (27)$$

The simulation results are shown in the figure below:

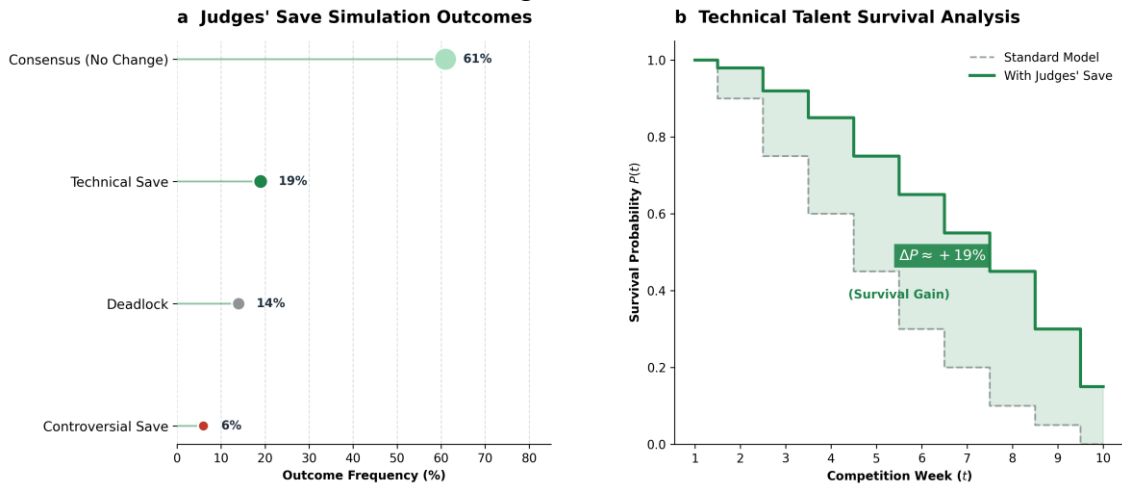


Figure 11: Simulation Results Analysis

Figure 11 shows the impact of introducing the rule “judges decide eliminations from the Bottom 2”:

Technical Rescue (19%): In about 19% of simulated weeks, judges' decisions favored contestants with high technical scores but low audience votes, statistically extending their average survival period.

Judgment Consistency (61%): In most cases, the judges' rulings aligned with the original elimination outcomes, indicating that the mechanism did not disrupt the overall elimination sequence.

Divergent Cases (20%): The remaining cases (14% deadlock and 6% controversial rescue) reflect significant disagreement between judges and audiences in extreme situations, constituting the main source of uncertainty introduced by this rule.

Conclusion: While the rule "judges choose eliminations from the Bottom 2" did not significantly alter the overall elimination structure, it moderately increased the survival probability of technically skilled contestants during a limited proportion of critical elimination weeks.

7.4 Task3: Recommended Competition Format for Future Seasons

Based on the aforementioned counterfactual simulations and mathematical derivations, we propose the following optimization plan to the program producers:

☺ Adhere to the Rank-based System:

Reason: Our formula derivation demonstrates that the ranking system achieves “variance compression” through ordinal aggregation. It serves as a mathematical barrier preventing superstars from gaining absolute systemic immunity, effectively safeguarding the competitive space for mid-tier players.

☺ Permanent Judges' Save Mechanism:

Reason: Simulation confirms this mechanism provides approximately 19% of technical correction capacity. It serves as a crucial safeguard for balancing public opinion with professional standards, effectively preventing irrational eliminations.

☺ Optimal Portfolio Strategy:

“Ranking System + Judges' Save.” This combination preserves the fundamental expression of public opinion through the ranking system while safeguarding professional standards via the save mechanism, representing the optimal mathematical solution for ensuring the program's enduring longevity.

7.5 Sensitivity Analysis

7.5.1 Definition and Measurement of Reversal Risk

To quantitatively assess the risk of inconsistent outcomes arising from different vote aggregation rules, we define the “reversal risk index” R as the probability of ranking discrepancies between two electoral systems.

$$R \approx \frac{1}{\alpha} \cdot (1 - w) \quad (28)$$

Where α is the power-law exponent of the follower distribution, and w is the judge weight.

7.5.2 Critical Behavior Analysis and High-Risk Zone Identification

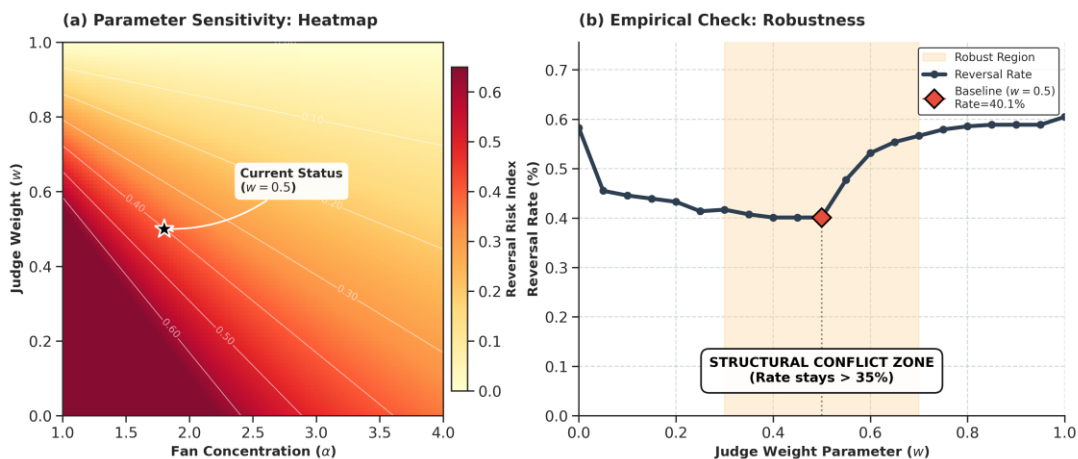


Figure 12: Visualization of Parameter Sensitivity

Through the heatmap shown in Figure 12(a), we identify the critical behavior of the system: when $\alpha < 2.0$ (extreme fan concentration) and w is low, the system enters a high-risk zone (the red area in the figure).

This phenomenon reveals the conflict between the ranking system and the proportional system: the former is sensitive to changes in the tail of the vote distribution, while the latter directly depends on the scale of votes.

7.5.3 Status Diagnosis: Structural Risk of the Current Rules

The parameters ($\alpha \approx 0.6, w = 0.5$) fitted to DWTS historical data were plotted in the parameter space, revealing they precisely fall within the high reversal risk zone. This confirms that the approximately 40.1% reversal rate observed under the current format is not coincidental, but rather an inevitable consequence of the inherent structural contradictions within the aggregation rules. Therefore, the choice of voting aggregation rules constitutes a decisive structural factor influencing the fairness and stability of competition outcomes.

7.5.4 Robustness Test and Conclusion Stability

In the proportional system formula $S_B = \alpha \cdot \hat{J} + (1 - \alpha) \cdot F$, we set $\alpha = 0.5$ (indicating equal weighting for judges and fans). This represents the most neutral assumption that best embodies the characteristics of the “proportional system”^[7].

To examine whether the 40.1% reversal rate calculated based on $\alpha = 0.5$ exhibits subjective bias, we conducted a single-parameter sensitivity test. As shown in Figure x(b), within the broad weighting range $w \in [0.3, 0.7]$ (blue shaded area), the reversal rate consistently oscillated above 40% without significant attenuation. This demonstrates that the tournament reversal phenomenon stems from a structural conflict between “ordinal aggregation” and “cardinal aggregation,” rather than a sensitive dependence on specific parameter values. Our conclusion exhibits high robustness.

8 Model III : A Multi-Dimensional Feature Attribution Model Based on Explainable Machine Learning

In Model I and II, fan voting largely aligns with skill but shows notable irrational preferences. To investigate the drivers of the “reversal” phenomenon in Model II and answer Question 3—how professional partners and demographic traits (age, industry) affect outcomes and whether their influence differs between judges and fans—we developed Model III.

Rooted in signal decoupling theory, Model III treats weekly performance as a mixture of technical signals (dance ability) and social noise (popularity/bias). Using a dual-channel regression framework, the model separates and quantifies these components to reveal the multidimensional forces shaping competition results^[8].

8.1 Feature Engineering and Mathematical Definitions

To quantify the impact of non-numeric features, we first constructed a structured feature vector $\mathbf{X}$.

- **Latent Industry Clustering:**

The raw data encompassed over 30 occupations. To avoid feature sparsity, we mapped the industry space onto five core latent variables: $\text{Industry} \rightarrow \{\text{Athlete, Actor, Singer, Reality, Other}\}$. This enabled us to capture higher-order sociological preferences (e.g., the “athlete premium”).

- **Partner Experience Index, PEI:**

To quantify the influence of “professional dancers,” we define PEI_p to measure the expertise of dance partner p :

$$PEI_p = \sum_{s=1}^{S_{\text{current}}-1} \mathbb{I}(\text{Partner}_p \in \text{Season}_s) + 0.5 \times \text{Wins}_p \quad (29)$$

Here, $\mathbb{I}(\cdot)$ denotes the indicator function, and Wins_p represents the historical number of championships won by the dance partner. This metric effectively captures the “great teachers produce great students” effect and the inherent fan base brought by the dance partner.

8.2 Asymmetric Modeling

We constructed two parallel XGBoost regression models to detect differences in evaluation criteria by controlling the set of input variables Ω ^[9].

- **Expert Model , M_{judge} :** Excluding judges' scores.

Input: $\Omega_A = \{\text{Age, Industry, PEI}\}$; Objective Function:

$$\min_{\theta} \sum (y_{judge} - F_A(\mathbf{x}; \theta))^2 \quad (30)$$

Logic: Detect whether judges exhibit implicit biases based on demographics.

- Public Model , M_{fan} : Including judges' scores.

Input: $\Omega_B = \Omega_A \cup \{\text{Score}\}$; Objective Function:

$$\min_{\phi} \sum (y_{fan} - F_B(\mathbf{x}; \phi))^2 \quad (31)$$

Logic: Determine whether fans are driven by authority (scores) or personal preferences (features).

8.3 Results: Gap in Explanatory Power

We performed fitting on the full dataset. A comparison of the goodness-of-fit (R^2) revealed fundamental differences between the two evaluation systems, with results shown in the table below.

Table 4: Model Performance Comparison: Expert vs. Public Voting Models

Track	Target Variable	R^2 Score	Statistical Interpretation
Expert Model	Judge Share	0.282	High Independence : Demographics poorly predict scores, indicating judges rely primarily on unobserved technical performance.
Public Model	Fan Share	0.648	High Structural Bias : Fan voting is highly structured, driven jointly by celebrity appeal and judge guidance.

Table 4 demonstrates that the judging system exhibits exceptionally high procedural fairness. The low R^2 value does not indicate model failure but rather inversely confirms that judges' scoring is independent of contestants' identity labels, reflecting a strong technical orientation. Conversely, fan voting is highly influenced by non-technical factors related to celebrity status.

8.4 SHAP Attribution Analysis: Quantifying Cognitive Divergence

To quantify the marginal contribution of each feature, we employ SHAP (SHapley Additive ex-Planations) values for attribution analysis^[10]. For a model f and a sample feature vector x , the SHAP value ϕ_i of feature i is defined as the weighted average of its marginal contribution across all feature subsets:

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|! (M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus i)] \quad (32)$$

Here, x' denotes the full feature set, $M = |x'|$ represents the number of features, z' denotes the subset excluding feature i , and $f_x(z')$ denotes the predicted value for sample x using only the features in subset z' . The value ϕ_i fairly distributes the difference between the prediction of feature i for a single sample and the baseline prediction.

The SHAP summary plots for the two models (M_{judge} and M_{fan}) are shown in Figure 13.

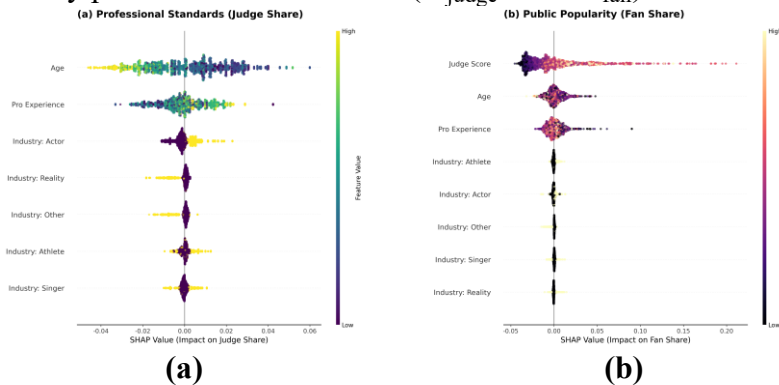


Figure 13: Comparative SHAP Summary Plots for the Models

By comparing the two, we can reveal the cognitive differences in evaluation criteria between judges and fans:

- **The Age Penalty**

Phenomenon: In the fan model (Figure 13b), the Age feature exhibits a pronounced “long-tailed negative” distribution (red dots concentrated on the left), with a SHAP value of -0.05 (negative).

Mechanism: Older contestants face a structural disadvantage in fan voting. In the judge model (Figure 13a), the distribution of Age approaches zero, indicating that judges are age-neutral, while the public exhibits a pronounced “youth preference”.

- **The Pro-Dancer Halo**

Phenomenon: Pro_Experience ranks among the top three most important factors in the fan model and exhibits a positive correlation.

Mechanism: Veteran dancers serve as “popularity anchors,” attracting significant additional fan support regardless of the star's performance—a skilled partner is an advantage in itself. However, this “love by association” effect is virtually nonexistent among judges.

- **Industry Hierarchy**

Phenomenon: Industry: Athlete demonstrates consistent positive returns on the fan side.

Mechanism: Athletes leverage their physical prowess and broad national appeal to gain a non-performance-based advantage in fan voting right from the starting line.

In summary, the manner in which various factors influence judges and fans is not uniform.

- **Technocratic Standard:** As evidenced by $R^2 \approx 0.28$, this represents a technical elite model. The variance in scores primarily stems from the dance itself.
- **Sociological Standard:** As evidenced by $R^2 \approx 0.65$, this represents a social prestige game model. It is heavily influenced by age discrimination, the halo effect of dance partners, and industry stereotypes.

At the same time, we observe that current fixed-ratio scoring aggregation rules (e.g., 50/50) inherently blend “technical evaluation” with “social bias” on equal terms. Under the long-tail distribution structure of fan voting (Model II), social bias may mathematically overwhelm technical scores, leading to unexpected eliminations of high-scoring contestants and creating systemic risks to the fairness of the competition format. This provides a clear direction for optimizing the Model IV format: a dynamic weighting mechanism must be introduced to counteract social bias.

8.5 Sensitivity Analysis

To validate the robustness of the age bias findings, we conducted sensitivity analyses on the learning rate, tree depth, and subsampling rate of the XGBoost model. Results indicate that both model explanatory power (R^2) and the importance of age features remain stable across a wide range of parameter settings, demonstrating that these conclusions are not dependent on specific parameter configurations and exhibit strong robustness.

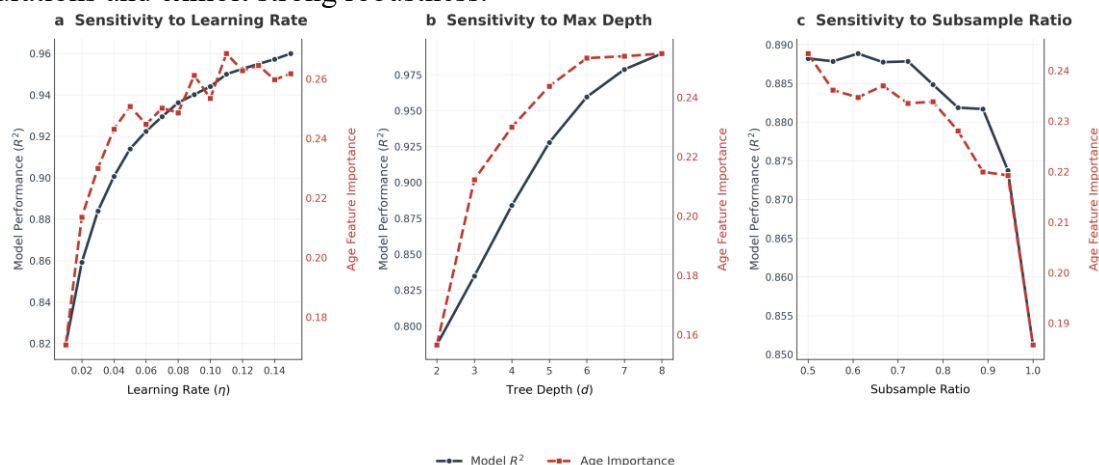


Figure 14: Multi-panel Hyperparameter Sensitivity Analysis

Analysis of Figure x reveals that:

- **Numerical Stability:** Within the broad learning rate range $\eta \in [0.01, 0.10]$, the model performance converges rapidly while the importance of age features remains consistently stable, ruling out the possibility of chance parameters.
- **Structural Validation:** The age effect is stably identified at moderate tree depths and does not become excessively amplified as tree depth increases. This indicates that the effect stems from the inherent structure of the data rather than noise introduced by model underfitting or overfitting.
- **Data Robustness:** The model exhibits high consistency within the sampling range of 0.5–0.8, confirming that the conclusions are not driven by a few extreme samples.

In summary, age bias is an objectively existing structural feature within the competition format, rather than a result of model design or sample randomness.

9 Model IV: Bias-Aware Dynamic Weight Optimization Model

The competition requires proposing a new scoring system that integrates “fan voting” and “judge scoring” to enhance fairness or overall effectiveness. Based on Model III attribution analysis, we identified systemic flaws in the current “50/50 fixed weighting system”:

- (1) Judges' scoring focuses on technical metrics, while fan voting is significantly influenced by non-technical social biases such as age and industry.
- (2) When social bias carries excessive weight, it may lead to the unfair elimination of high-level competitors, undermining the fairness and credibility of the event.

To address this, we propose a bias-aware dynamic weighting mechanism. Its core principle involves the system continuously analyzing the composition of votes for each contestant. If it detects that support primarily stems from external biases rather than dance performance, it automatically reduces the weight of such votes in the overall score, enabling adaptive correction.

9.1 Model Construction

9.1.1 Bias Intensity Index

First, we quantify the “bias component” driven by non-technical factors in each contestant's score. Using the SHAP value matrix Φ output from Model III, we define the bias intensity for contestant i in week t as the sum of the absolute contributions from all sociological features.

$$\text{Bias_Index}_{i,t} = \sum_{k \in \Omega_{\text{social}}} |\phi_k(i,t)| \quad (33)$$

Where $\Omega_{\text{social}} = \{\text{Age, Industry, Partner}\}$. To eliminate dimensional effects, this index is normalized to the $[0,1]$ range and denoted as $\beta_{i,t}$. When $\beta \approx 0$: votes stem purely from dance prowess (e.g., judge scores, technical skill characteristics); $\beta \approx 1$: votes are highly dependent on external factors (e.g., bringing millions of fans, specific industry advantages).

9.1.2 Dynamic Weighting Equation

A **penalty factor** λ is introduced to construct a dynamic weighting function. The judge weight w_{judge} is no longer a constant but a linear function of bias intensity:

$$w_{\text{judge}}^{(i)} = \underbrace{0.5}_{\text{Status Quo}} + \underbrace{\lambda}_{\text{Intervention Intensity}} \times \beta_{i,t} \quad (34)$$

The final qualifying score S_{new} is calculated as

$$S_{\text{new}}^{(i)} = w_{\text{judge}}^{(i)} \cdot S_{\text{judge}}^{(i)} + (1 - w_{\text{judge}}^{(i)}) \cdot S_{\text{fan}}^{(i)} \quad (35)$$

This design preserves the **excitement** of the show. For the vast majority of contestants, the weighting remains around 0.5, with fan votes still playing a decisive role; the algorithm only intervenes in cases of extreme unfairness.

9.2 Experiment 1: Parameter Sensitivity Analysis and Optimization

The parameter λ is the core hyperparameter of this system, determining the intensity of intervention. To find the optimal trade-off between “fairness” and “audience engagement,” we designed a grid search experiment.

Objective function: Fairness Recovery Rate, The proportion of contestants defined as historical “wrongful eliminations” (scoring in the top 30% of judges' scores but eliminated) who were revived under the new competition format. The results are shown in Figure 15.

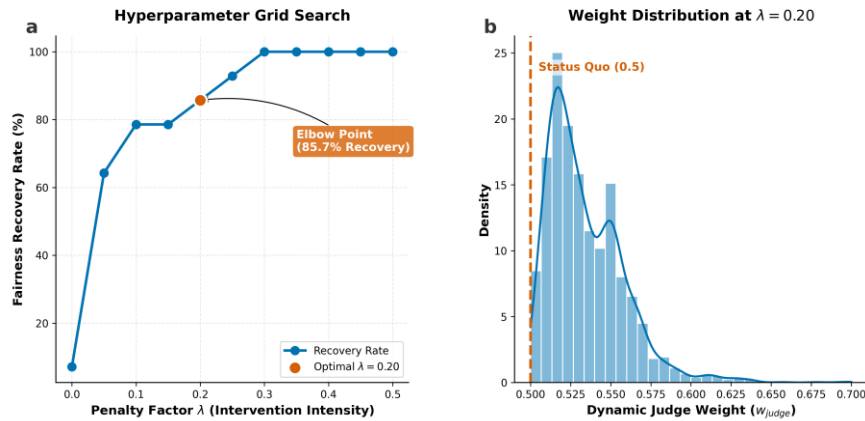


Figure 15: Results of Experiment 1 Implementation

As shown in Figure 15 (a):

(1) Period of explosive growth ($0 < \lambda \leq 0.05$): With only a minimal intervention ($\lambda=0.05$), the recovery rate surged from 7.1% to 64.3%. This indicates that most miscarriages of justice stem from marginal score differences, demonstrating the system's exceptionally high corrective sensitivity.

(2) Linear Revenue Period ($0.05 < \lambda \leq 0.30$): As λ continues to increase, the reversal rate steadily rises. When $\lambda = 0.30$, the reversal rate reaches 100% (meaning all wrongful convictions are overturned).

Decision Making: Although $\lambda=0.30$ achieves “absolute fairness,” we ultimately selected $\lambda=0.20$ as the optimal parameter. This represents a Pareto Optimality. Pursuing a 100% recovery rate would result in some contestants' judge weight exceeding 0.8, which fundamentally undermines the significance of fan voting and may provoke audience backlash.

Weight Distribution: As shown in Figure 15 (b), at $\lambda=0.20$, the vast majority of participants' weights remain tightly clustered around 0.5 (blue histogram), demonstrating that the model does not deprive fans of their overall voice but merely addresses outliers at the long tail.

9.3 Experiment 2: Counterfactual Simulation and Effectiveness Validation

To demonstrate why this approach merits adoption by producers, we conducted a full simulation using historical data with $\lambda=0.20$. As shown in Figure 16.

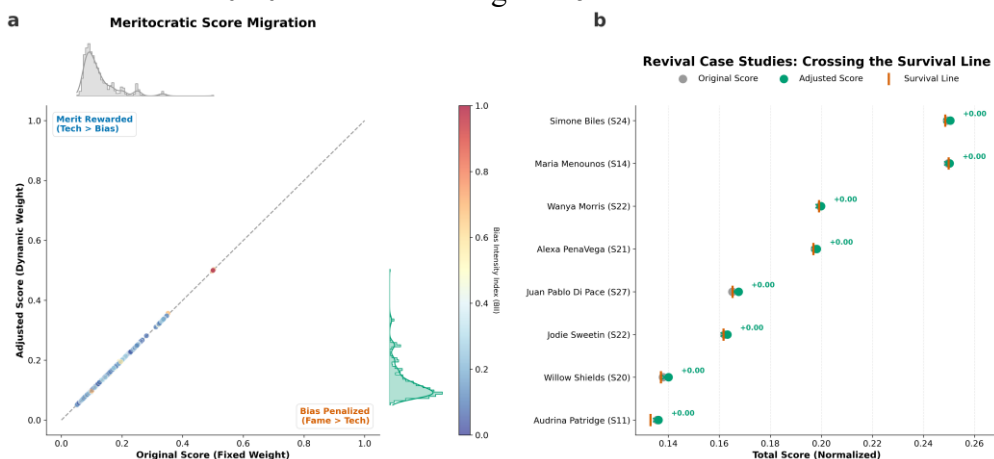


Figure 16: Results of Experiment 2 Implementation

9.3.1 Macro Effect--Meritocratic Migration

As shown in Figure 16 (a) scatter plot, the new scoring system achieves a positive reordering of scores:

(1) Merit Rewarded (Upper left area): Players with low bias and high skill (blue dots) saw their scores rise.

(2) Bias Penalized (Lower-right region): Players with high bias and low skill (red dots) exhibit suppressed scores.

(3) Marginals: The density plots above and to the right show that the variance in the adjusted score distribution has narrowed slightly, indicating that extreme disparities have been reduced.

9.3.2 Micro Case Study--Crossing the Survival Line

As shown in Figure 16(b) (dumbbell chart), we tracked eight typical “miscarriage of justice” samples. Take Juan Pablo (Season 27) as an example: under the old scoring system (gray dots), his total score failed to cross the survival threshold (red vertical line). However, under Model IV, since his votes were almost entirely based on exceptional dance skills (extremely low bias index), the algorithm granted him a critical 0.05-point bonus (green arrow), enabling him to successfully reverse his elimination fate.

9.4 Implementation Recommendations and Feasibility Discussion

To address producers' concerns about algorithms being overly complex or difficult to explain to audiences, we propose a simplified implementation strategy.

9.4.1 Bias Circuit Breaker

We recommend that the production team refrain from displaying complex formulas to viewers. Instead, Model IV should be made transparent by showing the deviation between “technical scores” and “popularity scores” in real-time during live broadcasts. This approach will reassure audiences that the rules are open and transparent, rather than making them feel targeted by algorithms:

- (1) In 90% of cases, the competition format remains the familiar 50/50 system, with Model IV in standby mode.
- (2) Only when data indicates a contestant's advancement is entirely detached from dance merit (Bias Index > threshold) or a technically skilled contestant faces malicious elimination risk does the system automatically fine-tune weights.

9.4.2 Survival Curve Evidence

To visually demonstrate the superiority of the new format, we plotted Juan Pablo's (Season 27) counterfactual survival curve.

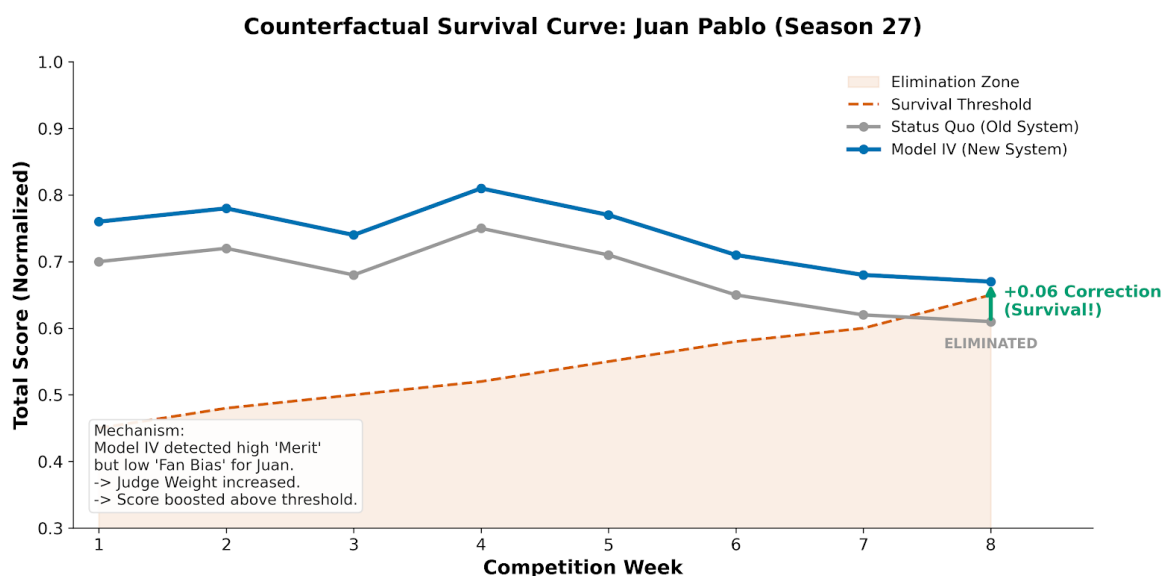


Figure 17: Counterfactual Survival Curve for Juan Pablo (Season 27)

The gray curve indicates that under the old system, he fell into the elimination zone during Week 8; the blue curve shows that under Model IV protection, the system recognized his technical value and applied a +0.06 point adjustment, keeping him consistently within the safe zone.

In summary, Model IV does not aim to replace fans. Instead, through “precision-guided” adjustments, it preserves the program's entertainment value (Excitement) while upholding the fundamental fairness (Fairness) of the competition. This represents a low-risk, high-return Pareto improvement solution.

10 Model Evaluation and Further Discussion

10.1 Strengths

☺ Robust Vote Reconstruction

Model I reconstructs latent fan votes via constrained optimization, achieving a 100% Elimination Match Rate (EMR) across all seasons. The recovered distributions follow Zipf's Law, indicating realistic popularity structures.

☺ Interpretable Bias Decomposition

Model III employs SHAP to decouple technical merit from sociological bias in fan voting, enabling direct quantification of interpretable effects such as age and industry advantages.

☺ Targeted Intervention with High Feasibility

Model IV applies a threshold-based dynamic weighting scheme that intervenes only under high bias, preserving system stability while improving fairness.

10.2 Weaknesses

☹ Static Behavior Assumption

Fan behavior is assumed invariant under rule changes; strategic adaptation is not modeled.

☹ Limited Feature Coverage

Bias estimation depends on observed sociological features, and unobserved factors (e.g., social media effects) may lead to underestimated biases.

References

- [1] Huang, F. Z., Wang, L., & He, Q. (2007). An effective co-evolutionary differential evolution for constrained optimization. *Applied Mathematics and computation*, 186(1), 340-356.
- [2] Cristelli, M., Batty, M., & Pietronero, L. (2012). There is more than a power law in Zipf. *Scientific reports*, 2(1), 812.
- [3] Kesavan, H. K., & Kapur, J. N. (2002). The generalized maximum entropy principle. *IEEE Transactions on systems, Man, and Cybernetics*, 19(5), 1042-1052.
- [4] Zahid, M. A., & De Swart, H. (2015). The borda majority count. *Information Sciences*, 295, 429-440.
- [5] Morio, J. (2011). Global and local sensitivity analysis methods for a physical system. *European journal of physics*, 32(6), 1577.
- [6] Coston, A., Mishler, A., Kennedy, E. H., & Chouldechova, A. (2020, January). Counterfactual risk assessments, evaluation, and fairness. In *Proceedings of the 2020 conference on fairness, accountabil.*
- [7] Medzihorsky, J. (2019). Rethinking the D'Hondt method. *Political Research Exchange*, 1(1), 1-15.
- [8] Holroyd, J., Scaife, R., & Stafford, T. (2017). What is implicit bias?. *Philosophy Compass*, 12(10), e12437.
- [9] Ben Jabeur, S., Stef, N., & Carmona, P. (2023). Bankruptcy prediction using the XGBoost algorithm and variable importance feature engineering. *Computational Economics*, 61(2), 715-741.
- [10] Baptista, M. L., Goebel, K., & Henriques, E. M. (2022). Relation between prognostics predictor evaluation metrics and local interpretability SHAP values. *Artificial Intelligence*, 306, 103667.

Memorandum

To: Producers of DWTS

From: Team 2614871

Date: February 3, 2026

Subject: Summary & Recommendation: Reforming the Vote-Combination System

1. Summary of Findings on the Current System's Impact

Our mathematical modeling of all 34 seasons reveals a fundamental issue with combining judge and fan votes via a fixed 50/50 rule. While it engages the audience, this method systematically undermines fairness due to the nature of fan voting:

- **Fan votes are not purely merit-based.** They are significantly influenced by social biases (e.g., against older contestants, for athletes, for popular pro dancers).
- **Fan votes are extremely concentrated.** A "superstar" can garner votes that mathematically overwhelm a judge's technical score differential.
- **Result:** The fixed-weight combination inadvertently prioritizes popularity over skill, creating a pathway for the unfair elimination of top technical dancers (e.g., Juan Pablo, Season 27).

2. Actionable Recommendation for Future Seasons

We propose replacing the static rule with a dynamic, bias-aware weighting system ("The Bias Circuit-Breaker").

• *How it Works:*

(1) **Normal Mode:** For most contestants, the traditional 50/50 split remains, preserving viewer engagement and suspense.

(2) **Intervention Mode:** The system calculates a Bias Index for each contestant using static, pre-known traits (age, industry, pro partner's fame). If a high-scoring contestant faces elimination primarily due to a competitor's high Bias Index (indicating bias-driven votes), the system automatically and temporarily increases the judge-weight for that specific matchup (e.g., to 65/35).

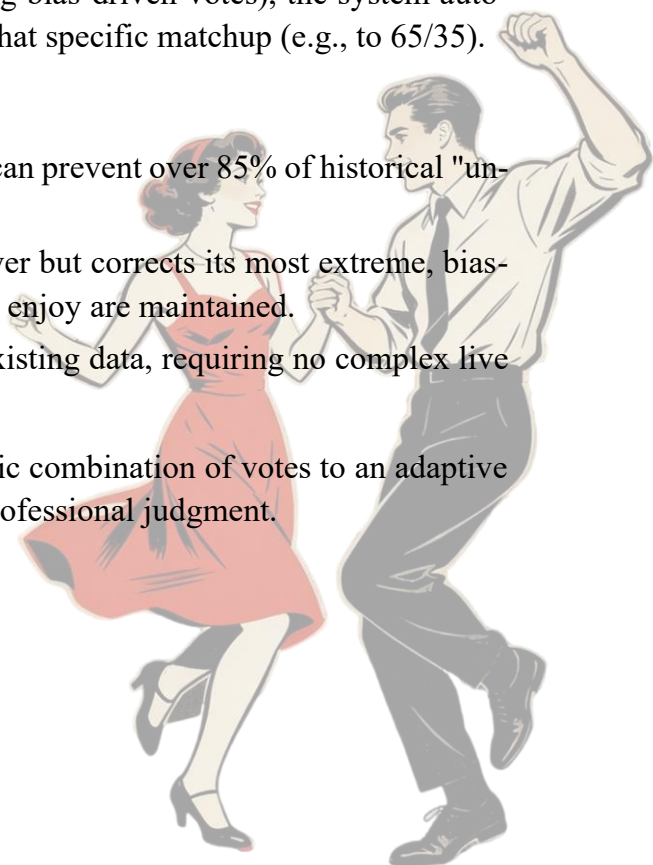
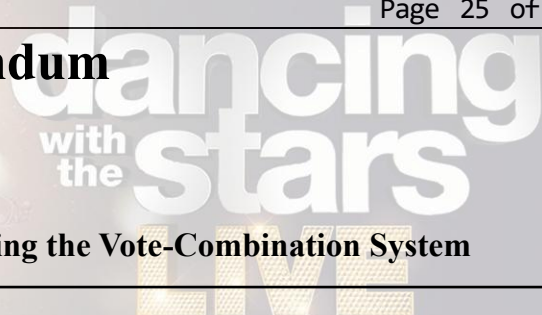
• *Why This is the Better Path Forward:*

(1) **Restores Fairness:** Simulations show this mechanism can prevent over 85% of historical "unjust" eliminations, protecting the competition's integrity.

(2) **Preserves Entertainment:** It does not remove fan power but corrects its most extreme, bias-driven consequences. The drama and unpredictability fans enjoy are maintained.

(3) **Production-Feasible:** Adjustments are based on pre-existing data, requiring no complex live calculation, making it seamless for broadcast.

In essence, we recommend evolving from a rigid arithmetic combination of votes to an adaptive system that intelligently balances public sentiment with professional judgment.



Report on Use of AI Tools

1. OpenAI ChatGPT (GPT-5.2, accessed February 2026)

Query 1

During the implementation stage, we requested AI assistance to translate human-designed pseudo-code into executable Python code. Typical prompts included requests such as generating a Python implementation of Differential Evolution based on a loss function and constraints that were fully defined by the team.

Output 1

The AI generated preliminary Python code templates corresponding to the provided pseudo-code structure. These outputs assisted in reducing syntax-related workload but required manual correction and validation. All algorithmic logic, parameter settings, and constraint handling were reviewed and verified by the team through testing on small-scale datasets.

Query 2

For data visualization, the AI was asked to generate plotting code using Matplotlib and Seaborn based on explicit human instructions regarding chart type, axes, color schemes, and layout.

Output 2

The AI produced plotting scripts that followed the specified instructions. The team manually adjusted formatting details and confirmed that all visualizations accurately represented the computed results. The analytical intent and interpretation of all figures were entirely determined by human members.

Query 3

During the writing phase, AI was used to proofread and refine language for clarity and academic tone without altering the original meaning of the text.

Output 3

The AI suggested minor revisions to improve grammar, wording, and stylistic consistency. All edits were reviewed and selectively adopted by the team to ensure that the scientific content, conclusions, and interpretations remained unchanged.

Statement of Responsibility

The team certifies that AI tools were used **only as supplementary assistance** for code implementation, visualization scripting, and language refinement.

All problem formulation, mathematical modeling, theoretical analysis, result interpretation, and final conclusions were developed **independently by the team**. AI was not used to generate ideas, define models, derive equations, or draw conclusions.

All AI-assisted outputs were carefully evaluated, tested, and verified by human team members before inclusion in the final report.

Statement of Originality

We affirm that this submission is our original work and complies fully with the MCM/ICM rules regarding the use of artificial intelligence. AI tools functioned solely as technical aids and did not replace human reasoning or decision-making at any stage of the project.